Chapter 5
Magnetostatics, Faraday's Law, Quasi-Static Fields

Diego Juarez.

These notes follow Jackson's SI notation. The magnetic-flux density is $\mathbf{B}$, the magnetic field in matter is $\mathbf{H}$, the vector potential is $\mathbf{A}$, the current density is $\mathbf{J}$, the magnetization is $\mathbf{M}$, and the permeability of free space is μ_0. The aim here is not to paraphrase the book mechanically, but to write the chapter in a way that makes the structure of the subject visible while keeping Jackson's notation and logic.

Prepared from the uploaded Chapter 5 material.

Contents

1 Introduction and Definitions **2**

2 Biot–Savart Law **3**

3 Differential Equations of Magnetostatics and Ampère's Law **3**

4 Vector Potential **4**

5 Circular Current Loop **5**

6 Localized Current Distribution and Magnetic Moment **5**

7 Force, Torque, and Energy of a Localized Current Distribution **6**

8 Macroscopic Equations; B and H **7**

9 Methods of Magnetostatic Boundary-Value Problems **8**

10 Uniformly Magnetized Sphere **9**

11 Faraday's Law **9**

12 Inductance and Flux Linkage **10**

13 Magnetic Energy **10**

14 Quasi-Static Fields **11**

Global framework of Chapter 5

Electrostatics began from charges as sources of a conservative field. Magnetostatics begins from a different fact: there are no free magnetic charges. The sources are currents, and the field equations are correspondingly rotational rather than purely potential in character. The first half of the chapter develops the steady magnetic field of prescribed current distributions; the second half introduces macroscopic media, Faraday induction, and the quasi-static limit in which time dependence is slow enough that full wave propagation is not yet the central issue.

1. Introduction and Definitions

The first conceptual break with electrostatics is immediate. In electrostatics one writes

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0},$$

and this already announces what the sources are. In magnetostatics the corresponding fundamental statement is not a source equation but the absence of isolated magnetic charge,

$$\nabla \cdot \mathbf{B} = 0.$$

Hence magnetic field lines do not begin or end; they form closed curves or extend to infinity in such a way that the net flux through any closed surface vanishes.

Jackson introduces $\mathbf{B}$ operationally by the torque on a magnetic dipole $\boldsymbol{\mu}$,

$$\mathbf{N} = \boldsymbol{\mu} \times \mathbf{B}.$$

This is the magnetic analogue of defining $\mathbf{E}$ by force on a test charge, except that here the basic probe is not a monopole but a dipole. Historically this is not a cosmetic difference. Magnetism was first encountered through compasses and permanent magnets, so the dipole entered before the field equations were fully understood.

A current density $\mathbf{J}(\mathbf{x})$ is the flow of charge per unit area per unit time. Charge conservation gives the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.$$

For steady currents, the charge density is time independent and therefore

$$\nabla \cdot \mathbf{J} = 0.$$

This divergence-free property is one of the basic structural facts of magnetostatics. It is what allows the field equations to simplify so beautifully later.

2. Biot–Savart Law

The experimental relation between steady currents and magnetic induction is the Biot–Savart law. For a filamentary current element $I\,d\mathbf{l}$ and an observation point displaced by $\mathbf{x}$ from the element,

$$d\mathbf{B} = \frac{\mu_0}{4\pi} I \frac{d\mathbf{l} \times \mathbf{x}}{|\mathbf{x}|^3}.$$

The $1/r^2$ character survives from Coulomb's law, but the vector structure is entirely different: the source element contributes perpendicular both to the current direction and to the line joining source and field point. This already tells us that magnetic fields naturally circulate around currents.

For an infinitely long straight wire carrying current I, direct integration gives

$$|\mathbf{B}| = \frac{\mu_0 I}{2\pi R},$$

where R is the perpendicular distance to the wire. The field lines are circles concentric with the wire. This is one of the central pictures of magnetostatics and will later reappear in Ampère's law as the simplest symmetry argument.

The force law on a current element in an external field is

$$d\mathbf{F} = I\,d\mathbf{l} \times \mathbf{B}.$$

Hence for a continuous current distribution,

$$\mathbf{F} = \int \mathbf{J}(\mathbf{x}) \times \mathbf{B}(\mathbf{x})\,d^3x, \qquad \mathbf{N} = \int \mathbf{x} \times [\mathbf{J}(\mathbf{x}) \times \mathbf{B}(\mathbf{x})]\,d^3x.$$

These formulas are more than appendices to the field law. They show that $\mathbf{B}$ is the field quantity whose physical content is fixed by magnetic forces and torques.

3. Differential Equations of Magnetostatics and Ampère's Law

The integral Biot–Savart expression for a general current density is

$$\mathbf{B}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{x}') \times \frac{\mathbf{x} - \mathbf{x}'}{|\mathbf{x} - \mathbf{x}'|^3}\,d^3x'.$$

Using

$$\frac{\mathbf{x} - \mathbf{x}'}{|\mathbf{x} - \mathbf{x}'|^3} = -\nabla \frac{1}{|\mathbf{x} - \mathbf{x}'|},$$

one may rewrite it as

$$\mathbf{B}(\mathbf{x}) = \frac{\mu_0}{4\pi} \nabla \times \int \frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}\,d^3x'.$$

From this form the divergence equation follows immediately:

$$\nabla \cdot \mathbf{B} = 0.$$

To obtain the curl, one uses the vector identity $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$ together with the Green-function relation

$$\nabla^2 \frac{1}{|\mathbf{x} - \mathbf{x}'|} = -4\pi\delta(\mathbf{x} - \mathbf{x}').$$

After an integration by parts, the term involving $\nabla \cdot \mathbf{J}$ vanishes in the steady state, yielding

$$\boldsymbol{\nabla} \times \mathbf{B} = \mu_0 \mathbf{J}.$$

Thus the basic differential equations of magnetostatics are

$$\boxed{\boldsymbol{\nabla} \times \mathbf{B} = \mu_0 \mathbf{J}, \qquad \nabla \cdot \mathbf{B} = 0.}$$

They are the magnetic counterparts of the electrostatic equations $\boldsymbol{\nabla} \times \mathbf{E} = 0$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$, but the roles of divergence and curl have been interchanged.

Integrating $\boldsymbol{\nabla} \times \mathbf{B} = \mu_0 \mathbf{J}$ over an open surface and using Stokes' theorem gives the integral form,

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{rmenc},$$

Ampère's law. In highly symmetric situations, this is the magnetic analogue of Gauss's law.

4. Vector Potential

Because $\nabla \cdot \mathbf{B} = 0$, one may introduce a vector potential $\mathbf{A}$ such that

$$\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}.$$

This is not a trick but a consequence of the topology of divergence-free fields in simply connected regions. Substituting into Ampère's law gives

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \mathbf{A}) = \mu_0 \mathbf{J},$$

or

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J}.$$

The freedom

$$\mathbf{A} \to \mathbf{A} + \nabla\Psi$$

leaves $\mathbf{B}$ unchanged and is called a gauge transformation. Jackson chooses the Coulomb gauge,

$$\nabla \cdot \mathbf{A} = 0,$$

for which each Cartesian component satisfies a Poisson equation,

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}.$$

The solution in unbounded space is therefore

$$\mathbf{A}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} \, d^3x'.$$

This is the magnetic counterpart of the electrostatic scalar-potential integral. The difference is not merely scalar versus vector. In electrostatics the potential is already the directly useful quantity. In magnetostatics the vector potential is partly auxiliary, but it becomes indispensable once induction and flux are discussed.

5. Circular Current Loop

A circular loop is the natural testing ground for the formalism because it is simple enough to solve but already rich enough to show the geometry of magnetic multipoles. For a loop of radius a in the x–y plane carrying current I, symmetry implies that only the azimuthal component of $\mathbf{A}$ survives:

$$\mathbf{A} = A_\phi(r, \theta) \, \hat{\boldsymbol{\phi}}.$$

The exact result can be expressed in terms of complete elliptic integrals. Jackson then develops a spherical-harmonic expansion to expose the structure more transparently. Far from the loop,

$$B_r = \frac{\mu_0}{2\pi} \frac{I\pi a^2 \cos\theta}{r^3}, \qquad B_\theta = \frac{\mu_0}{4\pi} \frac{I\pi a^2 \sin\theta}{r^3}.$$

This is precisely the field of a dipole. It is therefore natural to define the magnetic moment of a planar loop by

$$\mathbf{m} = I(\text{area}) \, \hat{\mathbf{n}},$$

so that for a circle,

$$m = I\pi a^2.$$

This result is basic enough that one should remember it geometrically rather than only algebraically: magnetic moment equals current times oriented area.

6. Localized Current Distribution and Magnetic Moment

For a localized steady current, expand the vector potential for $|\mathbf{x}| \gg |\mathbf{x}'|$:

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} = \frac{1}{|\mathbf{x}|} + \frac{\mathbf{x} \cdot \mathbf{x}'}{|\mathbf{x}|^3} + \cdots .$$

The monopole term vanishes because the current is localized and divergenceless. The first nonvanishing contribution is dipolar, and Jackson shows that it can be written in terms of the magnetic moment

$$\mathbf{m} = \frac{1}{2}\int \mathbf{x}' \times \mathbf{J}(\mathbf{x}')\, d^3x'.$$

Then

$$\mathbf{A}(\mathbf{x}) = \frac{\mu_0}{4\pi}\frac{\mathbf{m}\times\mathbf{x}}{|\mathbf{x}|^3},$$

and the magnetic induction is

$$\mathbf{B}(\mathbf{x}) = \frac{\mu_0}{4\pi}\left[\frac{3\mathbf{n}(\mathbf{n}\cdot\mathbf{m}) - \mathbf{m}}{r^3}\right], \qquad \mathbf{n} = \frac{\mathbf{x}}{r}.$$

So the asymptotic magnetic field of any localized steady current distribution is dipolar.

For a closed plane circuit one can rewrite the moment as

$$\mathbf{m} = \frac{I}{2}\oint \mathbf{x}\times d\mathbf{l},$$

which reduces to current times area. If the current arises from orbiting charges with a common charge-to-mass ratio, then

$$\mathbf{m} = \frac{q}{2M}\mathbf{L}.$$

This classical relation between angular momentum and magnetic moment is the ancestor of the quantum-mechanical story, even though it is not the full story for intrinsic spin.

7. Force, Torque, and Energy of a Localized Current Distribution

In a slowly varying external field one expands $\mathbf{B}$ around a convenient origin:

$$B_k(\mathbf{x}) = B_k(0) + \mathbf{x}\cdot\nabla B_k(0) + \cdots.$$

The leading nonvanishing term in the force is then

$$\mathbf{F} = \nabla(\mathbf{m}\cdot\mathbf{B}),$$

using $\nabla\cdot\mathbf{B} = 0$. The leading torque is

$$\mathbf{N} = \mathbf{m}\times\mathbf{B}.$$

Hence the potential energy of a magnetic dipole in an external field is

$$U = -\mathbf{m}\cdot\mathbf{B}.$$

This has exactly the same algebraic form as the electric dipole energy $-\cdot\mathbf{E}$, but one should not overdo the analogy. The magnetic dipole is produced by a current that must be maintained. The quantity above is the interaction energy relevant to the mechanical orientation of the dipole, not a

complete accounting of field energy for arbitrary time-dependent processes.

Jackson also emphasizes a subtle point of singular structure. To make the spherical volume average of the dipole field come out correctly, the point-dipole field must contain a delta-function term:

$$\mathbf{B}(\mathbf{x}) = \frac{\mu_0}{4\pi}\left[\frac{3\mathbf{n}(\mathbf{n}\cdot\mathbf{m}) - \mathbf{m}}{r^3} + \frac{8\pi}{3}\mathbf{m}\,\delta(\mathbf{x})\right].$$

This term matters in hyperfine structure and contact interactions. It is not optional bookkeeping.

8. Macroscopic Equations; B and H

In matter, microscopic currents from orbital motion and intrinsic magnetic moments fluctuate on atomic scales. Averaging over a macroscopically small but microscopically large region introduces the magnetization $\mathbf{M}(\mathbf{x})$, the magnetic dipole moment per unit volume. Its contribution to the vector potential is analogous to the polarization term in electrostatics. Jackson shows that magnetization contributes an effective current density

$$\mathbf{J}_M = \boldsymbol{\nabla} \times \mathbf{M}.$$

Thus the macroscopic Ampèrian equation becomes

$$\boldsymbol{\nabla} \times \mathbf{B} = \mu_0(\mathbf{J} + \boldsymbol{\nabla} \times \mathbf{M}).$$

This motivates the definition of the macroscopic field

$$\mathbf{H} = \frac{1}{\mu_0}\mathbf{B} - \mathbf{M},$$

for which the equations become

$$\boxed{\boldsymbol{\nabla} \times \mathbf{H} = \mathbf{J}, \qquad \nabla \cdot \mathbf{B} = 0.}$$

These are directly analogous to the electrostatic macroscopic equations

$$\nabla\cdot = \rho, \qquad \boldsymbol{\nabla} \times \mathbf{E} = 0,$$

with $\mathbf{H}$ playing the role of in the sense of a source-counting auxiliary field.

In linear isotropic media,

$$\mathbf{B} = \mu\mathbf{H}.$$

Here μ is the permeability. For diamagnets and paramagnets this relation is usually linear with μ only slightly different from μ_0. For ferromagnets the constitutive law is nonlinear and history-dependent, $\mathbf{B} = \mathbf{F}(\mathbf{H})$, and hysteresis becomes essential.

Boundary conditions

Across an interface, the normal component of $\mathbf{B}$ is continuous and the tangential component of $\mathbf{H}$ jumps by any surface current density $\mathbf{K}$:

$$(\mathbf{B}_2 - \mathbf{B}_1) \cdot \mathbf{n} = 0, \qquad \mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) = \mathbf{K}.$$

For linear media without surface current this becomes

$$B_{2n} = B_{1n}, \qquad \frac{\mathbf{B}_{2t}}{\mu_2} = \frac{\mathbf{B}_{1t}}{\mu_1}.$$

When $\mu_1 \gg \mu_2$, field lines of $\mathbf{H}$ meet the high-permeability material almost normally. This is the magnetic analogue of the electrostatic conductor limit.

9. Methods of Magnetostatic Boundary-Value Problems

Jackson organizes the practical methods into three broad classes.

First, the generally applicable method is the vector potential. If the constitutive relation is explicit and simple enough, one solves for $\mathbf{A}$ from

$$\boldsymbol{\nabla} \times \mathbf{H}[\boldsymbol{\nabla} \times \mathbf{A}] = \mathbf{J}.$$

In linear media with constant μ in each region, Coulomb gauge reduces the problem to Poisson equations for $\mathbf{A}$ with matching conditions across boundaries.

Second, in current-free regions one may introduce a magnetic scalar potential Φ_M by

$$\mathbf{H} = -\nabla \Phi_M.$$

In linear media this gives

$$\nabla \cdot (\mu \nabla \Phi_M) = 0,$$

and in piecewise uniform media each region satisfies Laplace's equation. This route makes the analogy with electrostatics very tight, especially near high-permeability boundaries.

Third, for hard ferromagnets with specified magnetization $\mathbf{M}$ and no free current, one may work either with an effective magnetic charge density

$$\rho_M = -\nabla \cdot \mathbf{M},$$

and surface density

$$\sigma_M = \mathbf{n} \cdot \mathbf{M},$$

or with the effective current picture

$$\mathbf{J}_M = \boldsymbol{\nabla} \times \mathbf{M}, \qquad \mathbf{K}_M = \mathbf{M} \times \mathbf{n}.$$

The two descriptions are mathematically equivalent when handled consistently, but Jackson warns not to mix formulas carelessly across discontinuities.

10. Uniformly Magnetized Sphere

A uniformly magnetized sphere is the classic example because it can be solved exactly and illustrates all three methods at once. Let

$$\mathbf{M} = M_0\hat{\mathbf{z}}.$$

Then the scalar-potential method shows that only the dipole term survives. Inside the sphere,

$$\mathbf{H}_{\text{in}} = -\frac{1}{3}\mathbf{M}, \qquad \mathbf{B}_{\text{in}} = \frac{2}{3}\mu_0\mathbf{M}.$$

Outside, the potential is that of a dipole with moment

$$\mathbf{m} = \frac{4\pi a^3}{3}\mathbf{M}.$$

This problem is worth remembering because it gives the demagnetizing field in its simplest exact form. The internal $\mathbf{H}$ opposes the magnetization, while $\mathbf{B}$ remains parallel to it. That distinction between $\mathbf{B}$ and $\mathbf{H}$ is often where intuition first begins to wobble, so the sphere is the right place to fix it.

11. Faraday's Law

The static equations cannot be the whole story because changing magnetic flux produces electric fields with nonzero circulation. The local law is

$$\boldsymbol{\nabla} \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$

and the integral form is

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{\mathrm{d}}{\mathrm{d}t}\int_S \mathbf{B} \cdot \mathbf{n}\, da.$$

This is Faraday's law. Its content is profound: time-dependent magnetic induction does not create charge, it creates vorticity in the electric field. Thus the field ceases to be globally representable by a single-valued scalar potential alone.

The minus sign is Lenz's law in differential form. The induced electromotive force opposes the change that produces it. At the level of field theory this sign is not an afterthought; it is what makes electromagnetic induction dynamically self-consistent and energetically stable.

Once $\partial\mathbf{B}/\partial t \neq 0$, the electrostatic decomposition $\mathbf{E} = -\nabla\Phi$ is incomplete. One must write

$$\mathbf{E} = -\nabla\Phi - \frac{\partial \mathbf{A}}{\partial t},$$

with the second term carrying the nonconservative part.

12. Inductance and Flux Linkage

For a circuit carrying current I, the magnetic flux through a surface bounded by the circuit is

$$\Phi_B = \int_S \mathbf{B} \cdot \mathbf{n}\, da.$$

In linear media the flux is proportional to current, and one defines the self-inductance by

$$\Phi_B = LI.$$

For several coupled circuits,

$$\Phi_i = \sum_j L_{ij} I_j,$$

where the L_{ij} are inductance coefficients. Faraday's law then gives the induced emf,

$$\mathcal{E}_i = -\sum_j L_{ij} \frac{\mathrm{d}I_j}{\mathrm{d}t}.$$

The reciprocity $L_{ij} = L_{ji}$ is one of the most useful general theorems of circuit theory and follows from the same symmetry that underlies Green-function reciprocity in electrostatics.

13. Magnetic Energy

In a linear medium, the work required to build the current distribution can be written in field form as

$$W = \frac{1}{2} \int \mathbf{J} \cdot \mathbf{A}\, d^3x = \frac{1}{2} \int \mathbf{B} \cdot \mathbf{H}\, d^3x.$$

In vacuum this reduces to

$$W = \frac{1}{2\mu_0} \int B^2\, d^3x.$$

This is the magnetic counterpart of the electrostatic energy density $\frac{1}{2}\varepsilon_0 E^2$, but the route to it is different because currents rather than charges are the direct sources.

For a linear inductor one gets the familiar lumped expression

$$W = \frac{1}{2} L I^2.$$

This formula is easy to memorize and even easier to misuse. Its validity depends on linearity and on using the quasistatic regime where radiation can be neglected.

14. Quasi-Static Fields

The quasi-static approximation is the intermediate regime between strict statics and full wave electrodynamics. The fields vary in time, but slowly enough that retardation across the system is negligible on the timescale of interest. One keeps induction effects, because $\partial \mathbf{B}/\partial t$ matters, but discards wave propagation as a dominant phenomenon.

There are two complementary limits. In the electroquasistatic limit, displacement effects are weak and the field is still mostly determined by instantaneous charge distributions. In the magnetoquasistatic limit, induction and conduction currents dominate, while the displacement current remains negligible. Dimensional comparison of characteristic size L, timescale T, and light-travel time L/c determines whether this approximation is self-consistent.

In conducting media, quasi-static evolution often leads to diffusion-type equations rather than wave equations. With Ohm's law $\mathbf{J} = \sigma \mathbf{E}$ and neglect of displacement current, Maxwell's equations imply magnetic diffusion on a characteristic time scale of order

$$\tau \sim \mu \sigma L^2.$$

This is the time over which currents rearrange and fields penetrate into conductors. It is one of the cleanest examples of how field theory interpolates continuously between statics and dynamics.

Closing synthesis

Chapter 5 changes the tone of the subject. In electrostatics one could often pretend that the field was merely the gradient of a scalar and that the difficult part was the boundary conditions. In magnetostatics that innocence disappears. The source is a flow, the field naturally curls around it, and the cleanest mathematical object is a vector potential carrying gauge freedom. Once matter is introduced, one must distinguish between the fundamental field $\mathbf{B}$ and the auxiliary field $\mathbf{H}$; once time dependence is admitted, even gently, Faraday's law forces a nonconservative electric field into the story.

The chapter therefore does two jobs at once. It completes the static theory of magnetic fields produced by currents and magnetized matter, and it opens the door to electrodynamics proper by showing exactly where strict statics fails and quasi-statics begins. In that sense Chapter 5 is the hinge between electrostatics and the full Maxwell theory.